

EXPERIMENT – 1: PROBABILITY THEORY

AIM

To compute the probability of each class in the **Breast Cancer Wisconsin Diagnostic Dataset** using Probability Theory.

ALGORITHM

1. Import the required libraries (`sklearn` and `pandas`).
 2. Load the Breast Cancer Wisconsin Diagnostic dataset using `load_breast_cancer()`.
 3. Convert the dataset into a Pandas DataFrame.
 4. Add the target column and map:
 - 0 → Malignant
 - 1 → Benign
 5. Count the number of samples in each class.
 6. Calculate the probability of each class.
 7. Display the class counts and probabilities.
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PROGRAM

```
from sklearn.datasets import load_breast_cancer
import pandas as pd

# Load dataset
data = load_breast_cancer()

# Create DataFrame
df = pd.DataFrame(data.data, columns=data.feature_names)

# Add target column
df["Diagnosis"] = data.target

# Convert 0 and 1 into labels
df["Diagnosis"] = df["Diagnosis"].map({
    0: "Malignant",
    1: "Benign"
})

# Count samples
print("Counts:")
print(df["Diagnosis"].value_counts())

# Calculate probabilities
print("\nProbabilities:")
print(df["Diagnosis"].value_counts(normalize=True))
```

DATASET

Dataset Name: Breast Cancer Wisconsin Diagnostic Dataset

- Source : Scikit-learn
 - Total Samples : **569**
 - Number of Features : **30**
 - Number of Classes : **2**
 - Class Labels :
 - **Benign**
 - **Malignant**
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OUTPUT

Counts:

Diagnosis
Benign 357
Malignant 212

Probabilities:

Diagnosis
Benign 0.627417
Malignant 0.372583

CALCULATION

Total Samples = **569**

Benign Samples = **357**

Malignant Samples = **212**

Probability of Benign

$P(\text{Benign}) = \frac{357}{569} = 0.627417$

Probability of Malignant

$P(\text{Malignant}) = \frac{212}{569} = 0.372583$

RESULT

Thus, the probability of a randomly selected sample belonging to the **Benign** class is **0.627417 (62.74%)**, and the probability of belonging to the **Malignant** class is **0.372583 (37.26%)**.